AP BIOLOGY TOPIC 8.3

Population Ecology: Growth Equations, J/S Curves & Life History

UNIT 8: ECOLOGY (10–15% EXAM WEIGHT)

SYI-1.G | SYI-1.H | DN/DT = R_MAX·N | LOGISTIC GROWTH

Credits: mitali120501@gmail.com

§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 8.3 (POPULATION ECOLOGY: GROWTH EQUATIONS, J/S CURVES & LIFE HISTORY)

CED Core Vocabulary

Population — A group of individuals of the same species living and interbreeding in the same geographic area at the same time.

2. Uniform — Evenly spaced due to territoriality (nesting penguins) or chemical inhibition (allelopathy in desert sagebrush).

Variables Defined — N = Current population size | t = Time | dN/dt = Population growth rate (individuals added per unit time)

Logistic Growth (S-Shaped / Sigmoidal Curve) — Occurs when limited resources slow growth as N approaches carrying capacity K .

3 Spatial Dispersion Patterns — 1. Clumped (Most Common): Individuals aggregate in patches around localized resources (water holes, nutrients), social herding/pack defense, or clumped offspring...

3. Random — Unpredictable spacing in homogeneous environments without strong attractions/repulsions (e.g., wind-blown dandelion seeds).

Exponential Growth (J-Shaped Curve) — Occurs when resources are unlimited / abundant and constraints (predators/disease) are absent (e.g., bacteria in fresh broth, invasive species entering a new...

§2. 1. POPULATION STRUCTURE, DISPERSION & BIRTH/DEATH EQUATION

Demographics

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

1. Population Structure, Dispersion & Birth/Death Equation ➔ **Population** ➔ **3 Spatial Dispersion Patterns**

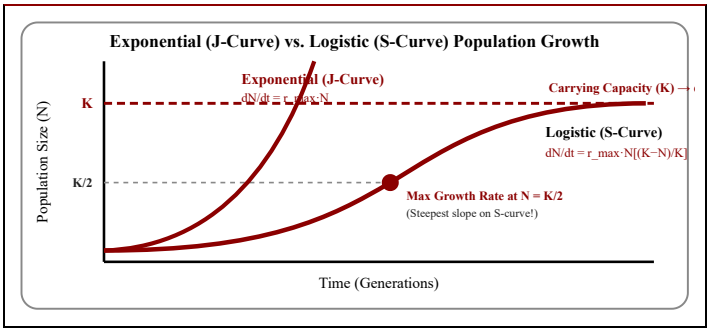
- **Population:** A group of individuals of the **same species** living and interbreeding in the same geographic area at the same time.
- **3 Spatial Dispersion Patterns:**
  - **1. Clumped (Most Common):** Individuals aggregate in patches around localized resources (water holes, nutrients), social herding/pack defense, or clumped offspring dispersal.
  - **2. Uniform:** Evenly spaced due to **territoriality** (nesting penguins) or chemical inhibition (allelopathy in desert sagebrush).
  - **3. Random:** Unpredictable spacing in homogeneous environments without strong attractions/repulsions (e.g., wind-blown dandelion seeds).
- **Fundamental Population Growth Equation (AP Formula Sheet):**

dN / dt = B − D (Change in Population Size over Time = Births − Deaths)

- If **B > D** → population grows (**dN/dt > 0**); if **B < D** → population declines (**dN/dt < 0**); if **B = D** → zero population growth (**dN/dt = 0**). *(In open populations, Immigration **+ I** and Emigration **− E** also contribute).*

§4. VISUALIZING EXPONENTIAL VS. LOGISTIC GROWTH CURVES

J-Curve vs. S-Curve



[EXAMPLE & BIOLOGICAL APPLICATION] **Worked AP Logistic Growth**

Calculation:

A rabbit population has **r_max = 0.5** *per month, carrying capacity* **K = 400**, *and current population size* **N = 300**. *Calculate* **dN/dt** *and the predicted population size after 1 month:*

- **dN/dt = 0.5 × 300 × [(400 − 300) / 400] = 150 × (100 / 400) = 150 × 0.25 = +37.5 ≈ +38 rabbits/month**

.

- **New Population Size (N_new): 300 + 38 = 338 rabbits**! *(Always check if the question asks for growth rate* **dN/dt** *or total new population size* **N + dN** *!)*

§5. 3. LIFE HISTORY STRATEGIES: R-SELECTED VS. K-SELECTED

Survivorship Trade-Offs

LIFE HISTORY TRAIT	*R*-SELECTED (OPPORTUNISTIC)	*K*-SELECTED (EQUILIBRIUM)
Typical Environment	Unstable, unpredictable, or disturbed habitats (low competition initially)	Stable, predictable habitats near carrying capacity **K** (intense competition)
Offspring Number & Size	**Many small offspring** per reproductive event; few survive	**Few large offspring**; high probability of surviving to adulthood
Maturation & Care	Early maturity, short lifespan, **little or no parental care** (often *semelparous* — 1 big brood)	Late maturity, long lifespan, **extensive parental investment** (*iteroparous* — repeated reproduction)
Survivorship Curve	**Type III Curve:** High juvenile mortality; survivors live long (insects, weeds, oysters, frogs)	**Type I Curve:** High survival in youth/middle age; mortality in old age (humans, elephants, whales)

§3. 2. EXPONENTIAL (J-CURVE) VS. LOGISTIC (S-CURVE) GROWTH

AP Formula Sheet Math

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

2. Exponential (J-Curve) vs. Logistic (S-Curve) Growth ➔ **Variables Defined** ➔ **Exponential Growth (J-Shaped Curve)**

Exponential: $dN/dt = r_{max} \cdot N$ | Logistic: $dN/dt = r_{max} \cdot N \cdot [(K - N) / K]$

- **Variables Defined:**
  - **N** = Current population size | **t** = Time | **dN/dt** = Population growth rate (individuals added per unit time)
  - **r_{max}** = Maximum per capita growth rate of population (intrinsic rate of increase under ideal conditions: **$r = (B - D)/N$**)
  - **K** = **Carrying Capacity** (maximum population size the environment can sustainably support)
- **Exponential Growth (J-Shaped Curve):**
  - Occurs when resources are **unlimited / abundant** and constraints (predators/disease) are absent (e.g., bacteria in fresh broth, invasive species entering a new habitat, or population rebounding after a bottleneck).
  - Notice: Per capita growth rate (**r_{max}**) stays constant, while overall population growth rate (**dN/dt**) accelerates continuously as **N** gets larger!
- **Logistic Growth (S-Shaped / Sigmoidal Curve):**
  - Occurs when **limited resources** slow growth as **N** approaches carrying capacity **K**.
  - The term **$(K - N) / K$** represents the fraction of **K** still available for growth:
    - When **N** is very small (**$N \ll K$**): **$(K-N)/K \approx 1$** → growth is nearly exponential.
    - **When $N = K / 2$ (Inflection Point):** Population growth rate **dN/dt** is at its **ABSOLUTE MAXIMUM**!
    - When **$N = K$**: **$(K-N)/K = 0$** → **$dN/dt = 0$** (population stabilizes at carrying capacity).
    - When **$N > K$** (Overshoot): **$(K-N)/K < 0$** → **dN/dt** becomes **negative**, and the population declines back toward **K**.



Support this site
Scan to donate

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 8.3 [MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]

Topic 8.3: Population Ecology: Growth Equations, J/S Curves & Life History ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 $SE_{\bar{x}}$)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 10, 11 & 12 (Energy Dynamics, Whole-Plant Transpiration & Pillbug Animal Behavior Choice Chambers): Choice chambers testing pillbug (Armadillidium vulgare) chemotaxis/hydrotaxis/kinesis (count pillbugs on wet vs. dry filter paper every 30 s for 10 min; evaluate using χ^2 against 50:50 null distribution). Potometer/gravimetric transpiration across light, wind, and humidity treatments normalized per cm^2 leaf surface area. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Ecological Equations: Population Growth Rate: $dN/dt = B - D$. Exponential Growth: $dN/dt = r_{max} \cdot N$ (J-curve). Logistic Growth: $dN/dt = r_{max} \cdot N \cdot [(K - N)/K]$ (S-curve; maximum instantaneous growth rate dN/dt occurs at $N = K/2$). Simpson's Diversity Index: $D_I = 1 - \Sigma(n / N)^2$ (ranges from 0 to 1; higher value = higher diversity). Net Primary Productivity: $NPP = GPP - R$. Trophic 10% Rule: ~10% of energy transfers to the next trophic level (~90% lost as metabolic heat/respiration). **Error Bar Rule (±2 $SE_{\bar{x}}$):** If the ±2 $SE_{\bar{x}}$ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Ecological Disruption & Trophic Cascade Case Study: Removal of a keystone predator (e.g., Pisaster ochraceus sea star in rocky intertidal zones, or sea otters in kelp forests) triggers a top-down trophic cascade: prey populations (mussels or sea urchins) explode unchecked, competitively excluding other invertebrates or overgrazing kelp producers, collapsing Simpson's Diversity Index. Similarly, agricultural N/P fertilizer runoff causes Cultural Eutrophication: algal blooms block light, aerobic bacteria decompose dead algae and deplete dissolved O_2, creating hypoxic dead zones.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 8.3 to prove why the claim holds.